



Course Name: Object Oriented Programming

Assignment Type: Complex Engineering Problem

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Pacman Console Game Report

Introduction

This project is a simple **Pacman-style console game** developed in Java. The game runs on a 5×5 grid where the player controls Pacman, collects food, and avoids a ghost.

Objective

- Move Pacman using keyboard controls
- Collect food to increase score
- Avoid the ghost that chases Pacman

- Win the game by reaching a score of 5
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Main Classes Used

1. pacman

- Stores position (x, y)
 - Handles movement:
 - `moveup()`
 - `movedown()`
 - `moveleft()`
 - `moveright()`
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2. ghost

- Stores ghost position
 - Uses a **chasing algorithm** instead of random movement
 - Moves toward Pacman using distance calculation
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3. food

- Stores position of food
 - Respawns at a random location after being eaten
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4. gameboard

- Creates a 2D grid (5×5)
- Displays:
 - `P` → Pacman
 - `G` → Ghost
 - `*` → Food
 - `-` → Empty space

Game Logic

Player Movement

- Controlled using:
 - **w** → Up
 - **s** → Down
 - **a** → Left
 - **d** → Right
- Movement is restricted within grid boundaries

Ghost Movement

- The ghost uses the **distance formula concept** to chase Pacman
- It checks all 4 directions and moves where distance is minimum

Formula used:

$$\text{distance} = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

Food System

- When Pacman reaches food:
 - Score increases by 1
 - New food appears at a random position

Game Conditions

Game Over:

- When ghost catches Pacman

Winning Condition:

- When score reaches 5
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Features

- Simple console-based gameplay
- Object-Oriented Programming (OOP)
- Intelligent ghost movement (AI behavior)
- Random food generation

ARCHITECTURE DIAGRAM

An architecture diagram is a graphical representation of the system structure that shows how different components like Pacman, Ghost, Food, and GameBoard interact with the main program to run the game.

